PART 2 (Continued): ASYMMETRIC WORKFLOW & DIGITAL CERTIFICATES

4. Asymmetric Message Exchange Scenario

Suppose **Alice** wants to send a confidential banking message to **Bob**:

1. Alice obtains Bob's public key from a public registry.
2. Alice encrypts her message using **Bob's Public Key**.
3. The ciphertext is transmitted across the open Internet. Even if an eavesdropper intercepts the transmission, they cannot read it because only Bob's private key can decrypt it.
4. Bob receives the ciphertext and decrypts it using his own **Bob's Private Key**.

5. Digital Signatures & Non-Repudiation — TB Pages 71–72

Asymmetric keys can also be used in reverse to authenticate the sender's identity and prove the document has not been tampered with:

- **Hashing:** The sender runs the message through a one-way mathematical hash function (e.g. SHA-256) to produce a fixed-length string called a *message digest*. If even a single character in the message changes, the hash changes completely.
- **Digital Signature Creation:** The sender encrypts the hash using their own **Private Key**. This encrypted hash is the *Digital Signature*.
- **Verification:** The recipient decrypts the signature using the sender's **Public Key** to reveal the original hash. They then compute their own hash of the received message. If both hashes match, two things are guaranteed:
  1. **Authentication & Non-Repudiation:** Only the sender could have created the signature because only they hold their private key.
  2. **Data Integrity:** The message was not modified in transit.

6. SSL/TLS & Digital Certificates — TB Pages 73–74

How does a user know a website's public key really belongs to them and not an imposter? Web browsers use **Digital Certificates** issued by trusted third-party **Certificate Authorities (CAs)** (such as DigiCert or Let's Encrypt).

A Digital Certificate contains:

- The organization / domain name (e.g., `kinderland.edu`).
- The website's Public Key.
- The Digital Signature of the Certificate Authority.
- The Validity period (Start and Expiry dates) and Serial number.

SSL/TLS Handshake: When connecting to a secure website (HTTPS), the browser uses asymmetric encryption to authenticate the server and securely exchange a temporary *session key*. Once established, high-speed symmetric encryption is used for the remainder of the session!